

SHYAM SARAN

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PROFESSIONAL SUMMARY

AI and Data Science graduate with hands-on experience developing localized, context-aware Generative AI systems. Proficient in building Retrieval-Augmented Generation (RAG) architectures using LangChain, FastAPI, and ChromaDB vector stores with custom LLM guardrails. Experienced in deploying full-stack data apps and eager to leverage predictive modeling and MLOps workflows to solve complex engineering challenges.

TECHNICAL SKILLS

Programming Languages:	Python, HTML, CSS
Libraries/Frameworks:	FastAPI, LangChain, NumPy, Pandas, Scikit-learn, Pytorch
Databases:	MySQL, ChromaDB
Tools & Platforms:	Power BI, Tableau, Git, GitHub, Docker, Jenkins (CI/CD)

EDUCATION

Bachelor of Technology in Artificial Intelligence and Data Science

SNS College of Engineering, Coimbatore (Sep 2021 – May 2025)

CGPA: 7.3

CERTIFICATIONS

- Data Science Master's Program – Besant Technologies
- Python Full Stack Development – A+ CADD Centre
- DSA in Python - Prepinsta

INTERNSHIP EXPERIENCE

Bdreamz Global Solutions | Data Science Intern (Sep 2025 – Feb 2026)

- Performed end-to-end data preprocessing and exploratory data analysis (EDA) on large structured datasets, applying missing value imputation, outlier detection, and feature engineering to ensure high-quality model-ready data
- Developed and evaluated multiple supervised machine learning models including Linear Regression, Polynomial Regression, Decision Tree, and Random Forest using Scikit-learn, achieving a best model R^2 score of 0.83 for a price prediction task
- Applied feature engineering techniques including one-hot encoding, categorical variable transformation, and train-test splitting, leveraging Pandas, NumPy, and Matplotlib/Seaborn for data manipulation and visualization

PROJECTS

OllamaChat | Python, FastAPI, LLM, LangChain

- Built a local AI chatbot using FastAPI, LangChain, Ollama, and RAG with ChromaDB vector store and PyMuPDF for PDF ingestion, enabling context-aware responses from custom knowledge bases
- Implemented real-time streaming responses via Server-Sent Events (SSE) with MongoDB persistence, in-memory session management, and LangChain RunnableWithMessageHistory for multi-turn conversations
- Designed a single-pass LLM guardrail classifier with LangChain prompt chaining and sentence-transformers embeddings, backed by a YAML-driven hot-reload configuration system for zero-downtime policy updates

Plate Pal | Python, Streamlit, scikit-learn, Pandas

- Built an AI-powered recipe recommendation app using cosine similarity and min-max normalisation to match personalised nutritional targets (calories, fat, carbs, protein) against a recipe dataset.
- Implemented health-aware filtering to restrict high-carb, high-fat, and high-calorie recipes for users with elevated blood sugar (≥ 126 mg/dL) or blood pressure (≥ 130 mmHg) using medical thresholds.
- Developed a full-stack Streamlit interface with session-state management, cached data loading, CSV export, and real-time nutritional insights for an optimised user experience.